Alex Morehead

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Machine Learning • Deep Learning • Computational Biology • High-Performance Computing

Education

University of Missouri — Columbia, Missouri
Ph.D. Computer Science — Aug 2020 - Present
- *Dissertation Advisor*: Prof. Jianlin Cheng
- *Cumulative GPA*: 4.0/4.0
- *Relevant Coursework*: Computational Intelligence, Machine Learning/Pattern Recognition, Deep Learning, NLP, Computer Vision, Unsupervised Learning, Computational Systems Biology, Design & Analysis of Algorithms

Missouri Western State University — St. Joseph, Missouri
B.S. Computer Science — Aug 2016 - May 2020
- *Minor*: Mathematics
- *Cumulative GPA*: 4.0/4.0 with General Studies Honors

Professional Experience

University of Missouri Bioinformatics & Machine Learning Lab — Columbia, Missouri
Graduate Research Assistant | Advisor: Jianlin Cheng — Aug 2020 – Present
- Research and develop novel geometric deep learning algorithms applicable to computational biology
- Authored and published at ICLR 2022 the Geometric Transformer architecture within DeepInteract, a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts
- Published the Geometry-Complete Perceptron Network model for representation learning of 3D molecular graphs

University of Missouri Allen Angel Capital Education Program — Columbia, Missouri
Investment Analyst | Advisor: William (W.D) Allen — Aug 2022 – Present
- Cultivate deal flow for identifying promising technology start-ups
- Pitched several technology companies to the fund, a subset of which have made it to due diligence

Absci AI Research Lab — New York City, New York
AI Scientist Intern | Manager: Joshua Meier — Jun 2022 – Aug 2022
- Researched and built deep learning algorithms for antibody design

Altec Information Services — Saint Joseph, Missouri
Software Development Intern | Advisor: Dan White — Aug 2018 – Aug 2020
- Reduced miscommunication between service centers globally by engineering over 5 new Angular web applications
- Built more than 6 secure backend APIs with the Spring framework in an agile development setting
- Collaborated closely with business stakeholders and analysts to understand data and problems needing to be solved

NSF Research Experience for Undergraduates in Data Science of Risk and Human Activity — Indianapolis, Indiana
Undergraduate Research Assistant | Program Directors: George Mohler, Mohammad Al Hasan — Jun 2019 - Aug 2019
- Authored an ensemble pipeline of 3 convolutional neural networks trained to detect gunshot sounds in the vicinity
- Deployed the pipeline to a cluster of Raspberry Pi 3 Model B+ microcomputers and evaluated its performance in real-world settings in collaboration with Indianapolis public safety officials
- Published as a manuscript and orally presented the gunshot sound detection project's results at the 2019 IEEE International Conference on Big Data

NSF Research Experience for Undergraduates in Synthetic Biology — St. Joseph, Missouri
Undergraduate Research Assistant | Program Directors: Todd Eckdahl, Jeffrey Poet — May 2018 - Aug 2018
- Investigated applications of combinatorial optimization to model lab experiments performed by synthetic biologists
- Produced Variant Sampler, a Java application for modeling the sample space of in vitro experiments
- Published results of the Variant Sampler project in the American Journal of Undergraduate Research

Center for Academic Support — St. Joseph, Missouri
Computer Science Content Tutor | Manager: Karen Luke — Feb 2017 - Aug 2018
- Tutored a total of 5 undergraduate students, 2 being from underrepresented groups
- Spent up to 1 hour with each student per week, reviewing data structures and object-oriented programming

Publications

• *Accepted/Published* •

Geometry-Complete Perceptron Networks for 3D Molecular Graphs — AAAI '23 – Graph Deep Learning Workshop, Washington D.C.
A. Morehead, J. Cheng

A Region-Based Deep Learning Approach to Automated Retail Checkout — CVPR '22 – AI City Workshop, New Orleans, LA
M. Shoman, A. Aboah, **A. Morehead**, Y. Duan, A. Daud, Y. Adu-Gyamfi

Geometric Transformers for Protein Interface Contact Prediction — ICLR '22, (Virtual)
A. Morehead, C. Chen, J. Cheng

High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function — IEEE SuperComputing '21, St. Louis, MO
M. Gao, P. L. Andersen, **A. Morehead**, S. Mahmud, C. Chen, X. Chen, N. Giri, R. Roy, F. Quadir, T. C. Effler, R. Prout, S. Abraham, W. Elwasif, J. Skolnick, J. Cheng, A. Sedova

Synthetic Biology Bicistronic Designs Support Gene Expression Equally Well in vitro and in vivo — AJUR '20, Journal
O. Koucky, J. Wagner, S. Aguilera, B. Bashaw, Q. Chen, A. Eckdahl, E. Edman, P. Gomez, N. Hanlan, N. Kempf, D. Mattoon, S. McKlin, C. Mazariegos, **A. Morehead**, S. Q. Ong, A. Peterson, M. Rojas, K. Roland, K. Schildknecht, H. Seligmann, K. Slater, A. Tauchen, R. Tittor, T. Travieso, D. Urban, C. Willis, J. Zhou, N. L. Snyder, L. J. Heyer, J. L. Poet, T. T. Eckdahl, A. M. Campbell

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi — IEEE BigData '19, Los Angeles, CA
A. Morehead, L. Ogden, G. Magee, R. Hosler, B. White, G. Mohler

• *Under Review* •

DRLComplex: Reconstruction of protein quaternary structures using deep reinforcement learning — arXiv '22, Under Review
E. Soltanikazemi, R. S. Roy, F. Quadir, N. Giri, **A. Morehead**, J. Cheng

DProQ: A Gated-Graph Transformer for Protein Complex Structure Assessment — arXiv '22, Under Review
X. Chen, **A. Morehead**, J. Liu, J. Cheng

EGR: Equivariant Graph Refinement and Assessment of 3D Protein Complex Structures — arXiv '22, Under Review
A. Morehead, X. Chen, T. Wu, J. Liu, J. Cheng

3D-Equivariant Graph Neural Networks for Protein Model Quality Assessment — bioRxiv '22, Under Review
C. Chen, X. Chen, **A. Morehead**, T. Wu, J. Cheng

• *Preprint* •

Semi-Supervised Graph Learning Meets Dimensionality Reduction — arXiv '22, Preprint Archive
A. Morehead, W. Chantapakul, J. Cheng

DIPS-Plus: The Enhanced Database of Interacting Protein Structures for Interface Prediction — arXiv '21, Preprint Archive
A. Morehead, C. Chen, A. Sedova, J. Cheng

Presentations

Neural Diffusion Models: Next-Generation Generative Deep Learning — University of Missouri – Deep Learning F22, Columbia, Missouri
A. Morehead

A Region-Based Deep Learning Approach to Automated Retail Checkout — CVPR '22 – AI City Workshop, New Orleans, Lousiana
A. Morehead, M. Shoman, A. Aboah

Geometric Transformers for Protein Interface Contact Prediction — ICLR '22, (Virtual)
A. Morehead

Geometric Transformers for Protein Interface Contact Prediction — Shandong University '22, Qingdao, Shandong Province, China (Virtual)
A. Morehead

High-Performance Toolbox for Deep Learning of Protein Structure and Function IEEE SuperComputing '21
A. Morehead, A. Sedova, M. Gao St. Louis, Missouri (Virtual)

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi IEEE BigData '19
A. Morehead, L. Ogden, G. Magee Los Angeles, California

Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi IUPUI Student Summer Poster Symposium
A. Morehead, L. Ogden, G. Magee Indianapolis, Indiana

Variant Sampling in vitro with a Scheduling Twist Alpha Chi National Convention '19
A. Morehead Cleveland, Ohio

Variant Sampling in vitro with a Scheduling Twist MWSU PORTAL Summer Research Showcase '18
A. Morehead St. Joseph, Missouri

Predicting Game Genres by Analyzing Code Structure CSCC Central Plains Conference '18
S. Frazier, **A. Morehead**, S. Prine Maryville, Missouri

Predicting Game Genres by Analyzing Code Structure MWSU Multidisciplinary Research Day '18
S. Frazier, **A. Morehead**, S. Prine St. Joseph, Missouri

Leadership Activities

ICLR Volunteer Apr 2022
NeurIPS Reviewer Sep 2021 – Present
Upsilon Pi Epsilon (ΥΠΕ) | University of Missouri Chapter Aug 2020 - Present
Alpha Chi (AX) | Missouri Western State University Chapter Mar 2018 – May 2020
Kappa Mu Epsilon (KME) | Missouri Western State University Chapter Mar 2018 – May 2020

Awards & Grants

CVPR Travel Grant *Jun 2022*
XSEDE Startup Hardware Grant *Dec 2021*
Dean's Engineering Excellence Fellowship *Aug 2020*
James W. and Joan M. O'Neill Graduate Fellowship in Engineering *Aug 2020*
MWSU Outstanding Graduating Computer Science Student Award *May 2020*
MWSU President's Honor Roll *May 2020*
Floyd Tesmer/Strayer University Prize in Computer Science and Engineering *Apr 2019*
Alpha Chi Region IV Scholarship *Apr 2019*
Grand Midwest Asynchronous Programming Contest 3rd Place Prize *Apr 2017*
East Side Lions Club Scholarship *May 2016*

Projects

PyTorch Lightning (20k+ Stars) | GitHub: https://github.com/Lightning-AI/lightning October 2022 – Present
- Work as a Lightning League Ambassador to ensure the quality of PyTorch Lightning code

BioPython (3k+ Stars) | GitHub: https://github.com/biopython/biopython Apr 2022 – Present
- Created functionality to remove large feature generation temporary files after all processing completes
- Collaborated with senior project developers to ensure proposed solution matched existing code standards

Graphein (300+ Stars) | GitHub: https://github.com/a-r-j/graphein Apr 2022 – Present
- Added workflow to catch elusive OS errors when executing external programs
- Worked with senior project developer to ensure workflow matched existing code standards

EGNN-PyTorch (150+ Stars) | GitHub: https://github.com/lucidrains/egnn-pytorch Apr 2021 – Present
- Noticed the omission of an important geometric unit test for this neural network architecture
- Developed and validated the code to test for node-wise permutation equivariance within the network

AMG-List (80+ Stars) | GitHub: https://github.com/amorehead/awesome-molecular-generation July 2022 – Present

- Curated an open-source list of machine learning research papers related to molecular generation
- Through pull requests, collaborate with other researchers to maintain the quality of papers presented

DeepInteract (40+ Stars) | GitHub: https://github.com/BioinfoMachineLearning/DeepInteract Oct 2021 – Present

- Designed and developed a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts
- Maintain project by addressing bug fixes, feature requests, and future planning

DIPS-Plus (20+ Stars) | GitHub: https://github.com/BioinfoMachineLearning/DIPS-Plus June 2021 – Present

- Curated (to date) the largest molecular dataset for machine learning of protein interface contacts
- Work to ensure ease of use and adoption by responding to feature requests and open issues

Technical Skills

Programming: Python 3 • Java • Angular 2+ • C/C++ • HTML/CSS • SQL • C# • R • MATLAB

Tools/Frameworks: PyTorch • TensorFlow • NumPy • Pandas • Spring Boot • NodeJS • Docker • AWS • Google Cloud

Methodologies: RESTFUL APIs • Version Control {Git} • Agile Development

News & Media Outreach

[1] MU Engineering. (2020, October 27). *Alex Morehead*. University of Missouri College of Engineering. https://engineering.missouri.edu/alex-morehead/.

[2] MU Engineering. (2021, October 18). *New protein prediction tool could accelerate biological discoveries*. University of Missouri College of Engineering. https://engineering.missouri.edu/new-protein-prediction-tool-could-accelerate-biological-discoveries/.

[3] ORNL. (2021, January 10). *Summit Powers Novel Protein Function Prediction Work*. Oak Ridge National Laboratory. https://www.ornl.gov/news/scientists-use-summit-supercomputer-deep-learning-predict-protein-functions-genome-scale/.

[4] HPCwire. (2021, January 13). *Summit Powers Novel Protein Function Prediction Work*. HPCwire. https://www.hpcwire.com/2022/01/13/summit-powers-novel-protein-function-prediction-work/.

[5] MarkTechPost. (2022, January 18). *Researchers Introduce High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function*. MarkTechPost Tech News – AI Paper Summary. https://www.marktechpost.com/2022/01/18/researchers-introduce-high-performance-deep-learning-toolbox-for-genome-scale-prediction-of-protein-structure-and-function/.